

# FROM SUPPRESSION TO REPAIR: MITIGATING HALLUCINATION IN LARGE VISION-LANGUAGE MODELS VIA LOCALIZED DISTRIBUTION ALIGNMENT

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## ABSTRACT

Object hallucination in large vision-language models (LVLMs) remains a major obstacle to trustworthy generation. An intuitive mitigation strategy is to suppress components of hidden representations associated with hallucinations. However, since these components may also contain useful visual-semantic information, directly suppressing them can impair the model’s multimodal capabilities. We therefore propose **ResOT**, a training-free inference-time method that repairs representations through localized distribution alignment. Specifically, ResOT confines the intervention to a low-dimensional residual subspace that captures hallucination-associated structures not explained by the dominant faithful subspace. Within this subspace, ResOT uses Gaussian optimal transport (OT) to define repair targets that align the hallucinated distribution with the faithful one while minimizing changes to the original representations. During inference, ResOT adaptively controls how far each token state moves toward its OT target. Across multiple benchmarks and three representative LVLMs, ResOT achieves the best overall performance in hallucination mitigation and captioning among the compared methods. It also improves multimodal performance over the corresponding vanilla LVLMs.

## 1 INTRODUCTION

Large vision-language models (LVLMs) have demonstrated strong performance in image captioning, visual question answering, and visual reasoning (Alayrac et al., 2022; Bai et al., 2023; Chen et al., 2024b). Powered by their robust multimodal alignment capabilities, LVLMs can generate fluent and informative responses given an input image. However, these responses are not always faithful to the image content. LVLMs may describe objects that do not appear in the image, a failure mode known as object hallucination (Bai et al., 2024; Li et al., 2023; Rohrbach et al., 2018). Such errors undermine the reliability of LVLMs and motivate research on hallucination mitigation.

One approach is to fine-tune LVLMs using additional data with hallucination annotations, but this alters pretrained model parameters and incurs substantial training costs (Jiang et al., 2024; Liu et al., 2024a). As a training-free alternative, recent work has explored inference-time interventions that mitigate hallucinations by directly modifying hidden representations (Liu et al., 2025; Lin et al., 2026). These methods commonly identify directions or subspaces associated with hallucinations and suppress the corresponding components (Yang et al., 2025a; Zhu et al., 2026; Dastmalchi et al., 2026). However, hallucination-associated signals may be entangled with useful visual-semantic information in hidden representations, making a clean separation difficult. Consequently, suppressing them may remove useful information and harm the model’s multimodal capabilities (Hua et al., 2025). This motivates a shift from suppression to repair: we aim to correct hallucination-associated deviations while preserving useful visual-semantic information encoded in the representation space.

Figure 1 provides a conceptual illustration of why suppression can be destructive. In many cases, a hallucinated response is not entirely incorrect: it may correctly describe the attributes and relations of objects in the image, but invent objects that are not present. This suggests that hallucinated representations may retain visual-semantic structure shared with faithful representations, resulting in partial overlap (Figure 1(a)). Applying suppression along a hallucination-associated direction can also affect faithful representations in the overlapping region, potentially discarding useful in-

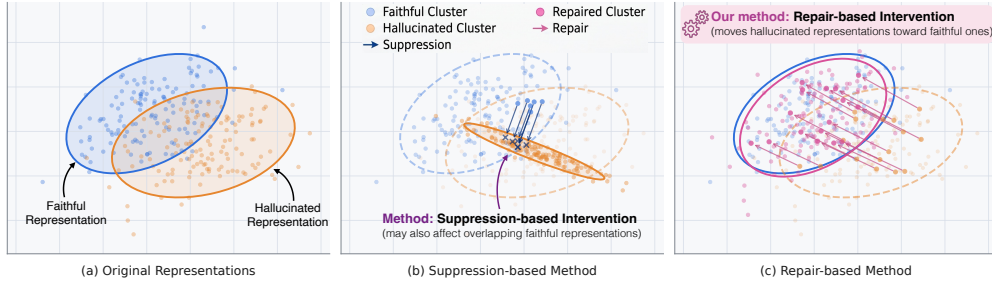


Figure 1: Conceptual comparison of suppression and repair. (a) Faithful and hallucinated representations form two distinct but partially overlapping clusters. (b) Suppression maps the hallucinated representations into a narrow band. The blue arrows show how faithful representations in the overlapping region are moved by the same operation. (c) A repair-based intervention moves hallucinated representations toward the faithful distribution while preserving the structure of the faithful cluster.

formation (Figure 1(b)). In contrast, a repair-based intervention moves hallucinated representations toward the faithful distribution while preserving the shared visual-semantic structure (Figure 1(c)).

In this paper, we propose **ResOT**, a training-free method that repairs representations during inference. Our method focuses on two questions: *where to repair* and *how to repair*. To answer the first question, ResOT identifies a low-dimensional residual subspace for repair. This subspace specifically captures deviations associated with hallucinations while excluding the dominant structures of faithful representations. Within this subspace, ResOT models the hallucinated and faithful distributions as Gaussians and derives a closed-form optimal transport (OT) map. Under the Gaussian approximation, this map defines a full repair target that moves the hallucinated distribution to the faithful one with minimal displacement. Because not every token state requires the full OT repair, a likelihood-based gate is used to adjust the repair strength at inference, assigning stronger repair to states more likely to be hallucinated. All updates are confined to the residual subspace, leaving the orthogonal components unchanged.

Our primary contribution is to shift the objective of intervention from potentially destructive suppression to localized distribution repair. ResOT realizes this objective by concentrating the repair where it is most needed while limiting its impact elsewhere. Experiments on three representative LVLMs demonstrate that ResOT substantially reduces hallucinations while achieving average relative gains of 32.18% in BLEU and 5.77% in MME over the corresponding vanilla models (Li et al., 2023; Rohrbach et al., 2018; Papineni et al., 2002; Fu et al., 2023). In our efficiency evaluation on LLaVA-1.5 (Liu et al., 2023), ResOT increases peak GPU memory usage by only 8.15 MiB while retaining 81.24% of the base model’s decoding speed.

## 2 RELATED WORK

### 2.1 HALLUCINATION IN LARGE VISION-LANGUAGE MODELS

Object hallucination may originate from various stages of LVLM processing. Weak visual perception, cross-modal misalignment, or overreliance on language prior can lead the model to generate content inconsistent with the image (Gu et al., 2025; Jain et al., 2024; Xing et al., 2024; Liu et al., 2024b). Training-based methods address these errors through supervised fine-tuning (Liu et al., 2024a), contrastive learning (Jiang et al., 2024), RLHF (Sun et al., 2024), or DPO (Yang et al., 2025b; Yu et al., 2024). However, these methods require additional supervision and parameter optimization, resulting in high training costs. Training-free methods, on the other hand, intervene during or after generation. Specifically, decoding-time approaches adjust token selections using visual contrast (Leng et al., 2024; Chen et al., 2024a), intermediate-layer predictions (Wang et al., 2025; Chuang et al., 2024), or attention signals (Zhou et al., 2025; Huang et al., 2024). Post-hoc approaches verify and revise unsupported content after generation (Zhou et al., 2024; Yin et al., 2024). Although these methods avoid retraining, they can introduce substantial inference overhead due to additional forward passes, complex decoding procedures, or external verification.

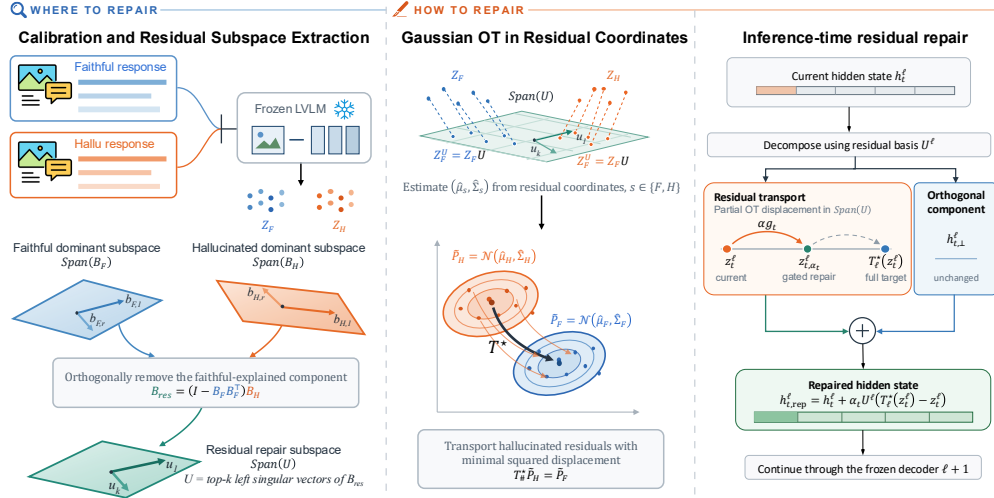


Figure 2: Overview of ResOT. Paired calibration representations identify a hallucination-dominant residual subspace after filtering out dominant faithful structure. A closed-form Gaussian OT map defines the minimum-displacement full repair target in this subspace. At inference, likelihood gating applies an adaptive fraction of the OT displacement, leaving the orthogonal component unchanged.

ResOT is most closely related to the representation-level interventions, which directly modifies hidden states. Existing methods steer latent activations (Liu et al., 2025; Li et al., 2025), intervene on attention heads (Chen et al., 2025), or suppress hallucination-associated components (Yang et al., 2025a; Zhu et al., 2026; Lin et al., 2026; Dastmalchi et al., 2026). However, a component associated with hallucination is not necessarily specific to it. The same component may also encode useful visual-semantic information. ResOT instead isolates hallucinated dominant residuals orthogonal to the dominant faithful structure and repairs their distribution through optimal transport. Thus, it differs from suppression in both where it intervenes and how it modifies the representation.

## 2.2 OPTIMAL TRANSPORT FOR REPRESENTATION ALIGNMENT

Optimal transport (OT) aligns two distributions by finding a transport plan that minimizes the cost of moving probability mass between them (Villani, 2009; Peyré & Cuturi, 2019). It has been widely used in feature alignment and domain adaptation (Courty et al., 2017; Damodaran et al., 2018). Empirical OT estimates the transport plan between the source and target samples, which can be expensive at scale. Sinkhorn iterations reduce this cost, but still rely on iterative optimization (Cuturi, 2013). In contrast, under squared Euclidean cost, the OT map between Gaussian distributions has a closed-form affine expression determined by their means and covariances (Peyré & Cuturi, 2019). ResOT uses this closed-form map to construct full repair targets in a low-dimensional residual subspace. A likelihood-based gate then controls how far each token state moves toward its target. Thus, ResOT does not require iterative OT optimization during generation.

## 3 METHOD

In this section, we first introduce the overall framework of ResOT. We then analyze its residual subspace construction and Gaussian OT repair, and compare them with projection-based suppression methods — the specific class of suppression methods considered in our theoretical analysis. Figure 2 provides an overview.

### 3.1 LVLM GENERATION AND CALIBRATION REPRESENTATIONS

LVLMs generally consist of a vision encoder, a vision-language projector, and a large language model (LLM) (Liu et al., 2023; Dai et al., 2023; Ye et al., 2024). Given an image  $\mathcal{V}$ , the vision encoder extracts visual features, and the projector maps them into the language space. The LLM

receives these features along with a text prompt  $x$  and generates a response token by token. At generation step  $t$ , let  $h_t^\ell \in \mathbb{R}^d$  denote the hidden state at layer  $\ell$  used to predict the next token  $y_t$ .

Like prior projection-based representation interventions (Yang et al., 2025a; Zhu et al., 2026), ResOT uses a paired calibration set to characterize the differences between faithful and hallucinated representations. The set contains  $N$  images; each image is paired with one faithful description and one hallucinated description. For each selected layer  $\ell$ , we feed both image–description pairs into the LVLM and collect the hidden states at the final text position. Stacking the faithful and hallucinated states separately gives  $Z_F^\ell, Z_H^\ell \in \mathbb{R}^{N \times d}$ . ResOT uses these matrices to construct a layer-specific residual subspace and estimate the corresponding Gaussian OT map. During inference, a likelihood-gated update based on this map repairs  $h_t^\ell$  before it is passed to subsequent layers. The LVLM parameters remain fixed during both calibration and inference.

### 3.2 RESIDUAL-SUBSPACE REPRESENTATION REPAIR

#### 3.2.1 LOCATING THE REPAIR SUBSPACE

ResOT confines repair to hallucination-dominant directions that are not explained by the dominant faithful subspace. Given  $Z_F, Z_H \in \mathbb{R}^{N \times d}$ , we omit the layer index  $\ell$  for clarity. Their dominant representation subspaces are obtained from

$$Z_F = L_F S_F R_F^\top, \quad Z_H = L_H S_H R_H^\top. \quad (1)$$

With the right singular vectors ordered by corresponding singular value, we retain the top- $c$  columns of  $R_F$  and  $R_H$  to form  $B_F = [b_{F,1}, b_{F,2}, \dots, b_{F,r}] \in \mathbb{R}^{d \times r}$  and  $B_H = [b_{H,1}, b_{H,2}, \dots, b_{H,r}] \in \mathbb{R}^{d \times r}$ , where  $B_F$  and  $B_H$  are orthonormal bases for the dominant faithful and hallucinated subspaces. The component of  $B_H$  explained by the faithful subspace is  $B_F B_F^\top B_H$ . Removing this component gives

$$B_{\text{res}} = B_H - B_F (B_F^\top B_H) = (I_d - B_F B_F^\top) B_H. \quad (2)$$

We compute the SVD  $B_{\text{res}} = \bar{L} \bar{S} \bar{R}^\top$  and let  $q = \text{rank}(B_{\text{res}})$ . For  $k \leq q$ , the residual basis is  $U = \bar{L}[:, 1 : k] \in \mathbb{R}^{d \times k}$ . Its columns are orthonormal and satisfy  $B_F^\top U = 0$ . The corresponding singular values measure how strongly the dominant hallucinated directions extend beyond the faithful subspace. All subsequent interventions are confined to  $\text{span}(U)$ .

#### 3.2.2 GAUSSIAN OPTIMAL TRANSPORT IN THE RESIDUAL SUBSPACE

Within the residual subspace, Gaussian OT defines a full repair target that aligns both the mean and covariance without removing residual directions. We project the calibration representations onto  $\text{span}(U)$ :  $Z_F^U = Z_F U$  and  $Z_H^U = Z_H U$ , where each row is a  $k$ -dimensional residual coordinate. Let  $(\hat{\mu}_F, \hat{\Sigma}_F)$  and  $(\hat{\mu}_H, \hat{\Sigma}_H)$  denote their empirical means and covariances. Finite calibration data may produce noisy or ill-conditioned covariance estimates. We therefore apply covariance shrinkage (Ledoit & Wolf, 2004) before computing the affine OT map and the Gaussian log-likelihoods used by the inference-time gate:

$$\hat{\Sigma}_s \leftarrow (1 - \rho) \hat{\Sigma}_s + \rho \frac{\text{tr}(\hat{\Sigma}_s)}{k} I_k + \epsilon I_k, \quad s \in \{F, H\}. \quad (3)$$

Here,  $\rho \in [0, 1]$  is the shrinkage coefficient and  $\epsilon > 0$  ensures positive definiteness. We then approximate the residual distributions as

$$\tilde{P}_F = \mathcal{N}(\hat{\mu}_F, \hat{\Sigma}_F), \quad \tilde{P}_H = \mathcal{N}(\hat{\mu}_H, \hat{\Sigma}_H). \quad (4)$$

We seek a transport map that aligns the hallucinated approximation with the faithful one while minimizing expected squared displacement:

$$T^* = \arg \min_{T \in \mathcal{T}} \mathbb{E}_{z \sim \tilde{P}_H} \|T(z) - z\|_2^2 \quad \text{s.t.} \quad T_\# \tilde{P}_H = \tilde{P}_F. \quad (5)$$

The constraint requires the transported Gaussian approximation to match the faithful Gaussian approximation. For non-degenerate Gaussian distributions, the optimal map has the closed-form affine solution (Brenier, 1991; Peyré & Cuturi, 2019):

$$T^*(z) = \hat{\mu}_F + A(z - \hat{\mu}_H), \quad (6)$$

where  $A = \hat{\Sigma}_H^{-\frac{1}{2}} \left( \hat{\Sigma}_H^{\frac{1}{2}} \hat{\Sigma}_F \hat{\Sigma}_H^{\frac{1}{2}} \right)^{\frac{1}{2}} \hat{\Sigma}_H^{-\frac{1}{2}}$ . Thus,  $T^*$  exactly transports  $\tilde{P}_H$  to  $\tilde{P}_F$ , i.e.,  $T_{\#}^* \tilde{P}_H = \tilde{P}_F$ , matching both the mean and covariance of the faithful Gaussian approximation. Among maps satisfying this constraint,  $T^*$  minimizes expected squared displacement and therefore defines the minimum-displacement full repair target in the residual subspace.

### 3.2.3 REPAIR AT INFERENCE

Each selected layer  $\ell$  has a residual basis  $U^\ell$  and an OT map  $T_\ell^*$ , both estimated offline. At decoding step  $t$ , the residual coordinate and orthogonal component are

$$z_t^\ell = (U^\ell)^\top h_t^\ell, \quad h_{t,\perp}^\ell = (I_d - U^\ell (U^\ell)^\top) h_t^\ell. \quad (7)$$

The full repair target at this layer is  $\tilde{z}_t^\ell = T_\ell^*(z_t^\ell)$ . However, the required repair strength varies across decoding states. ResOT therefore computes a lightweight gate from the pre-repair state at the first selected layer  $\ell_a$  and shares it across all selected layers:

$$s_t = \frac{1}{k} \left[ \log \mathcal{N}(z_t^{\ell_a}; \hat{\mu}_H^{\ell_a}, \hat{\Sigma}_H^{\ell_a}) - \log \mathcal{N}(z_t^{\ell_a}; \hat{\mu}_F^{\ell_a}, \hat{\Sigma}_F^{\ell_a}) \right], \quad g_t = \sigma(s_t). \quad (8)$$

A larger  $g_t$  indicates stronger relative evidence for the hallucinated approximation, so a larger fraction of the OT displacement is applied. We define the effective repair strength as  $\alpha_t = \alpha g_t$ , where  $\alpha \in [0, 1]$  controls the maximum displacement. We then interpolate between the original and transported residual coordinates:

$$z_{t,\alpha_t}^\ell = (1 - \alpha_t) z_t^\ell + \alpha_t \tilde{z}_t^\ell, \quad (9)$$

Combining this coordinate with the unchanged orthogonal component gives

$$h_{t,\text{rep}}^\ell = h_t^\ell + \alpha_t U^\ell (T_\ell^*(z_t^\ell) - z_t^\ell). \quad (10)$$

At inference, we cap the OT correction norm at a threshold estimated for each layer from hallucinated calibration states, preserving its direction and preventing unusually large updates. Thus, ResOT adaptively scales the OT displacement for each token within the residual subspace. The orthogonal component remains unchanged, and the repaired state propagates through subsequent frozen decoder layers.

### 3.2.4 THEORETICAL ANALYSIS

The following theory justify ResOT's residual subspace and Gaussian OT repair. First, we show that the residual basis maximizes alignment with the dominant hallucinated subspace while remaining orthogonal to the dominant faithful subspace. Second, we show that hard projection-based suppression incurs an unavoidable mismatch whenever it removes faithful energy. Finally, we establish that Gaussian OT is the minimum-displacement affine map among those matching the faithful mean and covariance, even for a non-Gaussian source distribution. All proofs are provided in Appendix A.

**Proposition 1.** Let  $B_{\text{res}} = (I_d - B_F B_F^\top) B_H$ , and  $U \in \mathbb{R}^{d \times k}$  contain its top- $k$  left singular vectors, where  $k \leq \text{rank}(B_{\text{res}})$ . For any orthonormal basis  $V \in \mathbb{R}^{d \times k}$  satisfying  $B_F^\top V = 0$ ,

$$\|V^\top B_H\|_F^2 \leq \|U^\top B_H\|_F^2 = \sum_{i=1}^k \sigma_i^2(B_{\text{res}}).$$

Thus, among all  $k$ -dimensional subspaces orthogonal to the dominant faithful subspace,  $\text{span}(U)$  has the greatest squared alignment with the dominant hallucinated basis.

**Proposition 2.** Let  $\tilde{P}_H, \tilde{P}_F \in \mathcal{P}_2(\mathbb{R}^k)$ , and let  $\mu_F$  and  $\Sigma_F$  be the mean and covariance of  $\tilde{P}_F$ . For  $T_\Pi(z) = \Pi z$ , where  $\Pi = \Pi^\top = \Pi^2$  is a rank-reducing orthogonal projection,

$$W_2^2((T_\Pi)_\# \tilde{P}_H, \tilde{P}_F) = W_2^2((T_\Pi)_\# \tilde{P}_H, (T_\Pi)_\# \tilde{P}_F) + \|(I_k - \Pi)\mu_F\|_2^2 + \text{tr}((I_k - \Pi)\Sigma_F).$$

The first term is the Wasserstein mismatch between the projected distributions. The last two terms equal the faithful energy removed by suppression. Therefore, whenever the faithful distribution has nonzero energy in the removed subspace, suppression leaves a strictly positive mismatch to the original faithful distribution, even if the projected distributions match exactly.

**Proposition 3.** Let  $Z \in \mathbb{R}^k$  be any random vector with finite second moment, mean  $\mu_H$  and covariance  $\Sigma_H \in \mathbb{S}_{++}^k$ . Given a target  $\mu_F \in \mathbb{R}^k$  and  $\Sigma_F \in \mathbb{S}_{++}^k$ , define

$$\mathcal{A}_{H \rightarrow F} = \{T_A(z) = \mu_F + A(z - \mu_H) \mid A\Sigma_H A^\top = \Sigma_F\}.$$

Then the Gaussian OT map  $T^*$  is the unique minimizer:

$$T^* = \arg \min_{T_A \in \mathcal{A}_{H \rightarrow F}} \mathbb{E} [\|T_A(Z) - Z\|_2^2].$$

When  $Z \sim \mathcal{N}(\mu_H, \Sigma_H)$  and the target is  $\mathcal{N}(\mu_F, \Sigma_F)$ ,  $T^*$  is the exact quadratic-cost OT map. For arbitrary  $Z$ , it still uniquely minimizes displacement among affine maps matching the target mean and covariance, but exact matching of a full target distribution is not generally guaranteed.

## 4 EXPERIMENTS

We organize our evaluation around three questions. **RQ1:** Does ResOT reduce object hallucination while preserving caption quality? **RQ2:** Does it retain general multimodal capability and open-ended generation quality? **RQ3:** Why does ResOT outperform direct suppression?

### 4.1 EXPERIMENTAL SETUP

**Baselines.** We compare ResOT against vanilla LVLMS and two categories of training-free mitigation methods. The vanilla baselines are LLaVA-1.5 (Liu et al., 2023), mPLUG-Owl2 (Ye et al., 2024), and InstructBLIP (Dai et al., 2023), all equipped with 7B language decoders and evaluated using greedy decoding. Decoding-time baselines include OPERA (Huang et al., 2024), VCD (Leng et al., 2024), HALC (Chen et al., 2024a), and LVLMS-Saliency (Zhang et al., 2026). Representation-level baselines include ICT (Chen et al., 2025), VTI (Liu et al., 2025), VISTA (Li et al., 2025), Nullu (Yang et al., 2025a), HulluEdit (Lin et al., 2026), and REVIS (Wu et al., 2026), which provide the most direct comparison to ResOT.

**Benchmarks and Metrics.** We evaluate ResOT along three dimensions: object hallucination, generation quality, and general multimodal capability. POPE (Li et al., 2023) measures discriminative object hallucination through yes/no object-existence questions. We report accuracy, precision, and F1 score. CHAIR (Rohrbach et al., 2018) evaluates hallucinated objects in open-ended captions using  $\text{CHAIR}_s$  ( $C_s$ ) and  $\text{CHAIR}_i$  ( $C_i$ ), while BLEU (Papineni et al., 2002) evaluates caption quality. MME (Fu et al., 2023) evaluates multimodal perception and cognition. We further use GPT-4o to assess the visual accuracy and detailedness of open-ended descriptions on sampled COCO images (Lin et al., 2014). Broader evaluations on MM-Vet, MMHal-Bench, and AMBER (Yu et al., 2023; Sun et al., 2024; Wang et al., 2023) are provided in Appendix C.1.

### 4.2 OBJECT HALLUCINATION AND CAPTION QUALITY

We evaluate ResOT in two complementary settings. POPE measures discriminative object hallucination, while CHAIR and BLEU assess hallucination and caption quality in open-ended generation.

**Results on POPE.** Table 1 reports results averaged over the random, popular, and adversarial splits. ResOT achieves the strongest overall performance across all three LVLM backbones. It achieves the highest accuracy and F1 score in every case, together with the highest precision on mPLUG-Owl2 and InstructBLIP. These improvements show that residual subspace repair effectively reduces object hallucination while maintaining accurate object recognition.

**Results on CHAIR.** Table 2 reports the COCO captioning results. Across all three LVLM backbones, ResOT achieves the lowest  $C_s$ ,  $C_i$  and the highest BLEU. This improvement demonstrates the central advantage of repair: ResOT improves caption faithfulness without sacrificing the visual-semantic information needed for high-quality generation.

### 4.3 GENERAL MULTIMODAL CAPABILITY AND OPEN-ENDED GENERATION

To answer RQ2, we evaluate whether ResOT preserves general multimodal capability and improves visual faithfulness without sacrificing detail in open-ended generation.

Table 1: Average POPE results across the random, popular and adversarial splits. Higher values indicate better performance. The best results are highlighted in bold.

| Method         | LLaVA-1.5       |                  |               | mPLUG-Owl2      |                  |               | InstructBLIP    |                  |               |
|----------------|-----------------|------------------|---------------|-----------------|------------------|---------------|-----------------|------------------|---------------|
|                | Acc. $\uparrow$ | Prec. $\uparrow$ | F1 $\uparrow$ | Acc. $\uparrow$ | Prec. $\uparrow$ | F1 $\uparrow$ | Acc. $\uparrow$ | Prec. $\uparrow$ | F1 $\uparrow$ |
| Vanilla        | 84.20           | 82.30            | 84.67         | 78.23           | 72.07            | 81.07         | 84.17           | 83.07            | 84.60         |
| OPERA          | 84.67           | 83.37            | 85.13         | 79.77           | 74.37            | 82.03         | 85.00           | 83.93            | 85.43         |
| VCD            | 81.13           | 77.83            | 82.30         | 76.73           | 70.83            | 79.77         | 75.97           | 78.20            | 81.63         |
| HALC           | 84.83           | 82.80            | 85.53         | 78.30           | 72.23            | 81.13         | 84.60           | 83.37            | 85.13         |
| VISTA          | 84.23           | 80.03            | 85.47         | 76.00           | 69.33            | 79.70         | 85.20           | 84.87            | 85.50         |
| ICT            | 85.43           | 83.10            | 86.07         | 79.07           | 73.37            | 81.57         | 55.50           | 53.00            | 69.20         |
| VTI            | 85.21           | 81.73            | 86.23         | 80.09           | 79.75            | 81.96         | 83.70           | 81.00            | 84.63         |
| Nullu          | 85.53           | <b>86.07</b>     | 85.70         | 77.87           | 71.60            | 80.87         | 84.63           | 83.20            | 85.20         |
| Revis          | 85.19           | 82.15            | 86.09         | 78.24           | 72.10            | 81.12         | 84.82           | 83.78            | 85.30         |
| HulluEdit      | 85.63           | 82.76            | 86.46         | 80.87           | 72.36            | 81.27         | 83.08           | 80.25            | 84.14         |
| LVLMS-Saliency | 85.12           | 81.96            | 86.05         | 78.01           | 71.78            | 80.98         | 84.46           | 83.09            | 85.03         |
| ResOT          | <b>86.13</b>    | 84.43            | <b>86.67</b>  | <b>82.50</b>    | <b>81.57</b>     | <b>83.60</b>  | <b>86.63</b>    | <b>86.93</b>     | <b>86.40</b>  |

Table 2: Captioning results on COCO evaluated by CHAIR and BLEU. Lower  $C_s$  and  $C_i$  indicate fewer hallucinated objects, while higher BLEU indicates better caption quality. The maximum generation length is set to 512 tokens. The best results are highlighted in bold.

| Method         | LLaVA-1.5        |                  |                 | mPLUG-Owl2       |                  |                 | InstructBLIP     |                  |                 |
|----------------|------------------|------------------|-----------------|------------------|------------------|-----------------|------------------|------------------|-----------------|
|                | $C_s \downarrow$ | $C_i \downarrow$ | BLEU $\uparrow$ | $C_s \downarrow$ | $C_i \downarrow$ | BLEU $\uparrow$ | $C_s \downarrow$ | $C_i \downarrow$ | BLEU $\uparrow$ |
| Vanilla        | 0.514            | 0.1451           | 0.1739          | 0.568            | 0.1639           | 0.1559          | 0.462            | 0.1310           | 0.1740          |
| OPERA          | 0.454            | 0.1317           | 0.1769          | 0.488            | 0.1503           | 0.1769          | 0.398            | 0.1242           | 0.1524          |
| VCD            | 0.484            | 0.1437           | 0.1743          | 0.580            | 0.1808           | 0.1689          | 0.566            | 0.1808           | 0.1698          |
| HALC           | 0.380            | 0.1270           | 0.1870          | 0.500            | 0.1432           | 0.1762          | 0.620            | 0.1770           | 0.1580          |
| VISTA          | 0.436            | 0.1336           | 0.1619          | 0.528            | 0.1638           | 0.1426          | 0.434            | 0.1345           | 0.1508          |
| ICT            | 0.498            | 0.1439           | 0.1797          | 0.716            | 0.2037           | 0.1585          | 0.482            | 0.1349           | 0.1196          |
| VTI            | 0.322            | 0.1195           | 0.2017          | 0.314            | 0.1194           | 0.2087          | 0.428            | 0.1236           | 0.1379          |
| Nullu          | 0.394            | 0.1150           | 0.1581          | 0.412            | 0.1395           | 0.1830          | 0.440            | 0.1429           | 0.1760          |
| Revis          | 0.430            | 0.1361           | 0.1740          | 0.460            | 0.1489           | 0.1711          | 0.440            | 0.1211           | 0.1744          |
| HulluEdit      | 0.404            | 0.1287           | 0.1669          | 0.362            | 0.1324           | 0.1515          | 0.442            | 0.1203           | 0.1633          |
| LVLMS-Saliency | 0.348            | 0.1132           | 0.2091          | 0.330            | 0.1188           | 0.1798          | 0.436            | 0.1288           | 0.1757          |
| ResOT          | <b>0.292</b>     | <b>0.0953</b>    | <b>0.2334</b>   | <b>0.258</b>     | <b>0.0898</b>    | <b>0.2504</b>   | <b>0.378</b>     | <b>0.1076</b>    | <b>0.1770</b>   |

**Results on MME.** Figure 3 reports the MME results. Compared with the vanilla models, ResOT improves the total, perception, and cognition scores across all three backbones. It achieves the highest total and perception scores on LLaVA-1.5 and mPLUG-Owl2, together with the best or tied-best results in most object related subcategories. These improvements show that ResOT improves visual faithfulness without compromising broader multimodal capability.

**GPT-4o Evaluation on COCO.** Following prior protocols (Wang et al., 2025; Yang et al., 2025a), we evaluate open-ended descriptions on the 500 COCO validation images selected by OPERA. We use the same evaluation prompt as prior work, repeat the scoring five times, and report the mean scores. We compare ResOT with the vanilla backbone and Nullu, a representative suppression-based method. As shown in Table 3, ResOT achieves higher accuracy than both baselines across all three backbones and higher detailedness in five of the six comparisons. These results demonstrate improved visual faithfulness while retaining informative content.

#### 4.4 REPRESENTATION ANALYSIS AND ABLATION STUDY

To answer RQ3, we examine whether the advantage of ResOT arises from both its intervention space and its repair operation. We visualize how the residual repair subspace exposes hallucination-related

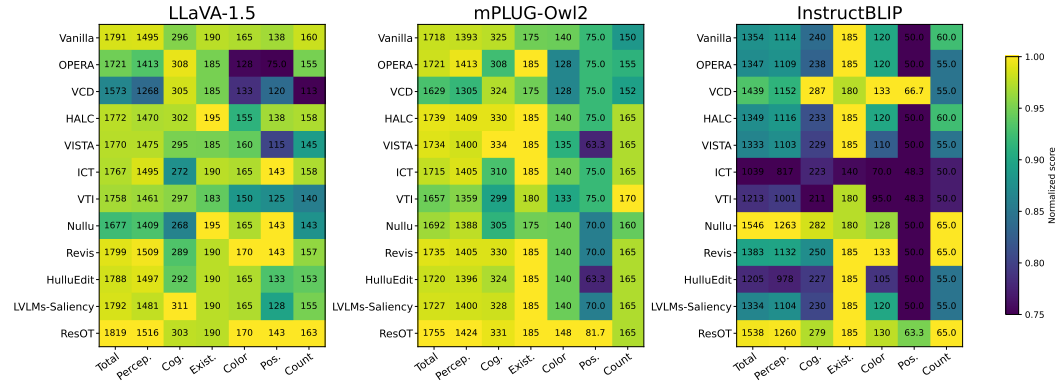


Figure 3: MME results across three LVLm backbones. Each cell reports the raw score, while color intensity indicates the normalized score within each metric. We report the total, perception and cognition scores, together with four perception subcategories: existence, color, position and count.

Table 3: GPT-4o evaluation on COCO image descriptions. Each entry reports the baseline/ResOT scores for visual accuracy (Acc.) and detailedness (Det.); higher values indicate better performance.

| Backbone     | Vanilla vs. ResOT  |                    | Nullu vs. ResOT    |                    |
|--------------|--------------------|--------------------|--------------------|--------------------|
|              | Acc.               | Det.               | Acc.               | Det.               |
| LLaVA-1.5    | 5.72 / <b>6.77</b> | <b>6.09</b> / 5.81 | 6.22 / <b>6.32</b> | 5.76 / <b>5.82</b> |
| mPLUG-Owl2   | 5.65 / <b>6.52</b> | 5.90 / <b>5.94</b> | 6.38 / <b>6.88</b> | 6.22 / <b>6.24</b> |
| InstructBLIP | 5.68 / <b>6.38</b> | 5.82 / <b>5.94</b> | 5.36 / <b>6.48</b> | 5.72 / <b>6.00</b> |

mismatch and evaluate calibration efficiency and robustness. Additional analyses of Gaussian fit and hyperparameter sensitivity are provided in Appendix D.

**Representation Distribution Visualization.** For a held-out analysis, we use 4000 calibration pairs to estimate the subspaces and repair parameters, while the remaining 1000 pairs for visualization. Figure 4 shows that faithful and hallucinated representations are highly mixed in the Nullu-style subspace. After removing components explained by the dominant faithful subspace, the residual repair subspace reveals a clearer distributional shift. Nevertheless, the two distributions still overlap because the residual subspace concentrates their dominant mismatch rather than separating them completely, and its orthogonality to the dominant faithful basis does not make all faithful residual coordinates vanish. Direct suppression would remove components also present in faithful representations along with hallucination-related deviations. ResOT instead uses a likelihood-gated OT update to move hallucinated representations toward the faithful distribution without projecting away the residual subspace. The visualization thus shows both how ResOT localizes the mismatch and why repair is preferable to suppression.

**Ablation Study.** We ablate the intervention space, repair operation, and adaptive gate. For subspace construction, we compare random, Nullu-style, and residual subspaces using Gaussian OT without gating. For the repair operation, we fix the residual subspace and compare direct suppression, mean-only alignment, and Gaussian OT, all without gating. Finally, we compare Gaussian OT with and without the gate in the residual subspace. Table 4 validates all three components. With Gaussian OT fixed, the residual subspace achieves the best hallucination, captioning, and MME results among the three subspace constructions. Within

Table 4: Ablation results on subspace construction and intervention operations.

| Variant              | $C_s \downarrow$ | BLEU $\uparrow$ | MME $\uparrow$ |
|----------------------|------------------|-----------------|----------------|
| RandSub + OT         | 0.516            | 0.1748          | 1787.17        |
| NulluSub + OT        | 0.424            | 0.1681          | 1764.82        |
| ResSub + Sup         | 0.378            | 0.1903          | 1784.96        |
| ResSub + Mean        | 0.320            | 0.2140          | 1798.35        |
| ResSub + OT w/o Gate | <b>0.292</b>     | 0.2268          | 1814.17        |
| ResSub + OT w/ Gate  | <b>0.292</b>     | <b>0.2334</b>   | <b>1819.02</b> |



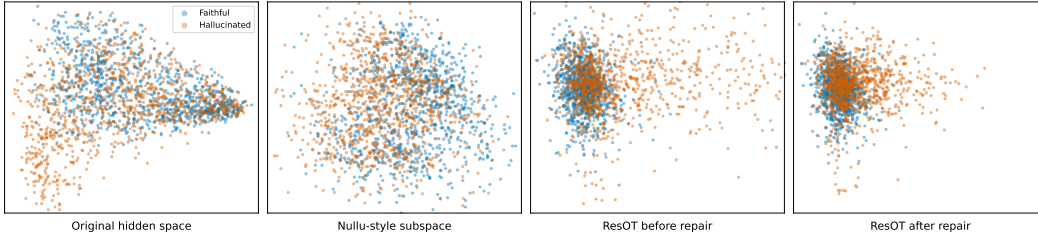


Figure 4: Projection-space visualization of faithful, hallucinated and repaired representations at layer 31. The three panels show the Nullu suppressed space, the ResOT residual subspace before repair, and the ResOT residual subspace after Gaussian OT repair.

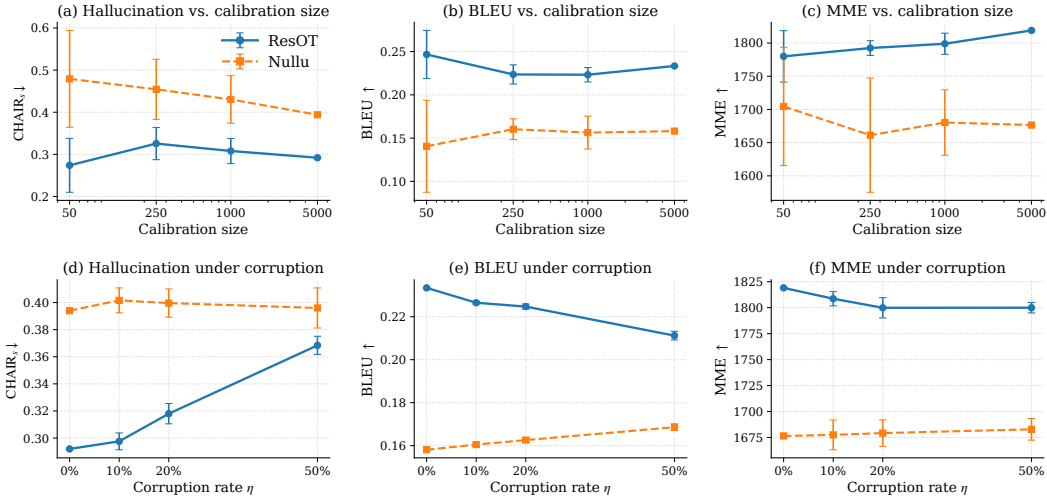


Figure 5: Effect of calibration size and corruption. We compare ResOT and Nullu under different calibration sizes and corruption rates. Error bars denote the standard deviation across random trials.

the residual subspace, Gaussian OT consistently outperforms suppression and mean-only alignment, confirming the benefit of covariance alignment beyond mean correction. The adaptive gate further increases BLEU and MME while retaining the same  $C_s$ . These results show that the residual subspace localizes where to repair, Gaussian OT determines how to repair, and the gate limits unnecessary changes.

**Calibration Set Analysis.** We re-estimate ResOT using nested calibration subsets of different sizes. To evaluate robustness to imperfect calibration, we corrupt a fraction  $\eta$  of the calibration pairs through image-text mismatches and faithful/hallucinated label swaps. Figure 5 shows that ResOT establishes a clear advantage with only 50 calibration pairs, achieving lower  $C_s$  and higher BLEU and MME than Nullu. Additional calibration data primarily improves estimation stability. ResOT also retains its advantage on all three metrics under corruption rates up to 50%, with its MME score remaining close to 1800. These results demonstrate that ResOT is highly efficient in calibration and robust to imperfect calibration data.

## 5 CONCLUSION

**ResOT** establishes a precise intervention principle for hallucination mitigation: operate only within a localized residual space and only to the degree indicated by each token’s hallucination tendency. Gaussian optimal transport realizes this principle as a minimum-displacement repair. And theoretical results characterizing the optimality of both localization and transport. Consistent empirical gains further establish localized repair as an effective alternative to direction suppression.

## REFERENCES

- Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35:23716–23736, 2022.
- Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023.
- Zechen Bai, Pichao Wang, Tianjun Xiao, Tong He, Zongbo Han, Zheng Zhang, and Mike Zheng Shou. Hallucination of multimodal large language models: A survey. *arXiv preprint arXiv:2404.18930*, 2024.
- Yann Brenier. Polar factorization and monotone rearrangement of vector-valued functions. *Communications on Pure and Applied Mathematics*, 44(4):375–417, 1991. doi: 10.1002/cpa.3160440402.
- Junzhe Chen, Tianshu Zhang, Shiyu Huang, Yuwei Niu, Linfeng Zhang, Lijie Wen, and Xuming Hu. Ict: Image-object cross-level trusted intervention for mitigating object hallucination in large vision-language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 4209–4221, 2025.
- Zhaorun Chen, Zhuokai Zhao, Hongyin Luo, Huaxiu Yao, Bo Li, and Jiawei Zhou. Halc: Object hallucination reduction via adaptive focal-contrast decoding. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pp. 7824–7846, 2024a.
- Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, et al. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 24185–24198, 2024b.
- Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James R Glass, and Pengcheng He. Dola: Decoding by contrasting layers improves factuality in large language models. In *International Conference on Learning Representations*, volume 2024, pp. 54158–54183, 2024.
- Nicolas Courty, Rémi Flamary, Devis Tuia, and Alain Rakotomamonjy. Optimal transport for domain adaptation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(9):1853–1865, 2017.
- Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, 2013.
- Wenliang Dai, Junnan Li, Dongxu Li, Anthony Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning. *Advances in neural information processing systems*, 36:49250–49267, 2023.
- Bharath Bhushan Damodaran, Benjamin Kellenberger, Rémi Flamary, Devis Tuia, and Nicolas Courty. Deepjdot: Deep joint distribution optimal transport for unsupervised domain adaptation. In *Proceedings of the European Conference on Computer Vision*, pp. 467–483, 2018.
- Hamidreza Dastmalchi, Aijun An, Ali Cheraghian, and Hamed Barzamini. Fighting hallucinations with counterfactuals: Diffusion-guided perturbations for lvlm hallucination suppression. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, et al. Mme: A comprehensive evaluation benchmark for multimodal large language models. *arXiv preprint arXiv:2306.13394*, 2023.
- Quanquan Gu, Weitong Zhang, Yihe Deng, and Linxi Zhao. Mitigating object hallucination in large vision-language models via image-grounded guidance. In *ICML 2025*, 2025.

- Zhenglin Hua, Jinghan He, Zijun Yao, Tianxu Han, Haiyun Guo, Yuheng Jia, and Junfeng Fang. Steering lvlms via sparse autoencoder for hallucination mitigation. *arXiv preprint arXiv:2505.16146*, 2025.
- Qidong Huang, Xiaoyi Dong, Pan Zhang, Bin Wang, Conghui He, Jiaqi Wang, Dahua Lin, Weiming Zhang, and Nenghai Yu. Opera: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 13418–13427, 2024.
- Jitesh Jain, Jianwei Yang, and Humphrey Shi. Vcoder: Versatile vision encoders for multimodal large language models. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 27992–28002. IEEE, 2024.
- Chaoya Jiang, Haiyang Xu, Mengfan Dong, Jiaxing Chen, Wei Ye, Ming Yan, Qinghao Ye, Ji Zhang, Fei Huang, and Shikun Zhang. Hallucination augmented contrastive learning for multimodal large language model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 27036–27046, 2024.
- Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.
- Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 13872–13882, 2024.
- Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, pp. 292–305, 2023.
- Zhuowei Li, Haizhou Shi, Yunhe Gao, Di Liu, Zhenting Wang, Yuxiao Chen, Ting Liu, Long Zhao, Hao Wang, and Dimitris N. Metaxas. The hidden life of tokens: Reducing hallucination of large vision-language models via visual information steering. In *International Conference on Machine Learning*, pp. 35799–35819, 2025.
- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *European conference on computer vision*, pp. 740–755. Springer, 2014.
- Yangguang Lin, Quan Fang, Yufei Li, Jiachen Sun, Junyu Gao, and Jitao Sang. Hulluedit: Single-pass evidence-consistent subspace editing for mitigating hallucinations in large vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11086–11095, 2026.
- Fuxiao Liu, Kevin Lin, Linjie Li, Jianfeng Wang, Yaser Yacoob, and Lijuan Wang. Mitigating hallucination in large multi-modal models via robust instruction tuning. In *International Conference on Learning Representations*, volume 2024, pp. 57689–57733, 2024a.
- Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- Sheng Liu, Haotian Ye, and James Y Zou. Reducing hallucinations in large vision-language models via latent space steering. In *International Conference on Learning Representations*, volume 2025, pp. 72402–72419, 2025.
- Shi Liu, Kecheng Zheng, and Wei Chen. Paying more attention to image: A training-free method for alleviating hallucination in lvlms. In *European Conference on Computer Vision*, pp. 125–140. Springer, 2024b.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pp. 311–318, 2002.

- Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019.
- Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. Object hallucination in image captioning. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 4035–4045, 2018.
- Zhiqing Sun, Sheng Shen, Shengcao Cao, Haotian Liu, Chunyuan Li, Yikang Shen, Chuang Gan, Liangyan Gui, Yu-Xiong Wang, Yiming Yang, et al. Aligning large multimodal models with factually augmented rlhf. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 13088–13110, 2024.
- Cédric Villani. *Optimal Transport: Old and New*, volume 338. Springer, 2009.
- Chenxi Wang, Xiang Chen, Ningyu Zhang, Bozhong Tian, Haoming Xu, Shumin Deng, and Huajun Chen. Mllm can see? dynamic correction decoding for hallucination mitigation. In *International Conference on Learning Representations*, volume 2025, pp. 13712–13736, 2025.
- Junyang Wang, Yuhang Wang, Guohai Xu, Jing Zhang, Yukai Gu, Haitao Jia, Jiaqi Wang, Haiyang Xu, Ming Yan, Ji Zhang, et al. Amber: An llm-free multi-dimensional benchmark for mllms hallucination evaluation. *arXiv preprint arXiv:2311.07397*, 2023.
- Jialin Wu, Wei Shi, Han Shen, Peigui Qi, Kunsheng Tang, Zhicong Huang, Binghao Wang, and Zhou Yang. Revis: Sparse latent steering to mitigate object hallucination in large vision-language models. *arXiv preprint arXiv:2602.11824*, 2026.
- Yun Xing, Yiheng Li, Ivan Laptev, and Shijian Lu. Mitigating object hallucination via concentric causal attention. *Advances in neural information processing systems*, 37:92012–92035, 2024.
- Le Yang, Ziwei Zheng, Boxu Chen, Zhengyu Zhao, Chenhao Lin, and Chao Shen. Nullu: Mitigating object hallucinations in large vision-language models via halluspace projection. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 14635–14645, 2025a.
- Zhihe Yang, Xufang Luo, Dongqi Han, Yunjian Xu, and Dongsheng Li. Mitigating hallucinations in large vision-language models via dpo: On-policy data hold the key. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10610–10620, 2025b.
- Qinghao Ye, Haiyang Xu, Jiabo Ye, Ming Yan, Anwen Hu, Haowei Liu, Qi Qian, Ji Zhang, and Fei Huang. mplug-owl2: Revolutionizing multi-modal large language model with modality collaboration. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 13040–13051, 2024.
- Shukang Yin, Chaoyou Fu, Sirui Zhao, Tong Xu, Hao Wang, Dianbo Sui, Yunhang Shen, Ke Li, Xing Sun, and Enhong Chen. Woodpecker: Hallucination correction for multimodal large language models. *Science China Information Sciences*, 67(12):220105, 2024.
- Tianyu Yu, Yuan Yao, Haoye Zhang, Taiwen He, Yifeng Han, Ganqu Cui, Jinyi Hu, Zhiyuan Liu, Hai-Tao Zheng, Maosong Sun, and Tat-Seng Chua. Rlhf-v: Towards trustworthy mllms via behavior alignment from fine-grained correctional human feedback. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 13807–13816, 2024.
- Weihao Yu, Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Xinchao Wang, and Lijuan Wang. Mm-vet: Evaluating large multimodal models for integrated capabilities. *arXiv preprint arXiv:2308.02490*, 2023.
- Xiaofeng Zhang, Yuanchao Zhu, Chaochen Gu, Xiaosong Yuan, Qian Zhao, Jiawei Cao, Feilong Tang, Sinan Fan, Yaomin Shen, Chen Shen, et al. Hallucination begins where saliency drops. *arXiv preprint arXiv:2601.20279*, 2026.
- Guanyu Zhou, Yibo Yan, Xin Zou, Kun Wang, Aiwei Liu, and Xuming Hu. Mitigating modality prior-induced hallucinations in multimodal large language models via deciphering attention causality. In *International Conference on Learning Representations*, 2025.

Yiyang Zhou, Chenhang Cui, Jaehong Yoon, Linjun Zhang, Zhun Deng, Chelsea Finn, Mohit Bansal, and Huaxiu Yao. Analyzing and mitigating object hallucination in large vision-language models. In *International Conference on Learning Representations*, volume 2024, pp. 56969–56998, 2024.

Xingyu Zhu, Junfeng Fang, Shuo Wang, Beier Zhu, Zhicai Wang, Yonghui Yang, and Xiangnan He. Mitigating hallucinations in large vision-language models without performance degradation. *arXiv preprint arXiv:2604.20366*, 2026.

## A PROOF

### A.1 PROOF OF PROPOSITION 1

Since  $B_F$  has orthonormal columns,

$$B_F^\top B_{\text{res}} = B_F^\top (I_d - B_F B_F^\top) B_H = 0.$$

Hence, the left singular vectors of  $B_{\text{res}}$  associated with nonzero singular values are orthogonal to  $\text{Span}(B_F)$ . Because  $k \leq \text{rank}(B_{\text{res}})$ ,  $B_F^\top U = 0$ .

For any feasible  $V$ ,  $B_F^\top V = 0$  implies  $V^\top B_F B_F^\top = 0$ . Thus,

$$V^\top B_H = V^\top (I_d - B_F B_F^\top) B_H = V^\top B_{\text{res}}.$$

Let  $B_{\text{res}} = Q \Sigma R^\top$  be a full singular value decomposition, where  $Q = [q_1, \dots, q_d]$  and  $\sigma_1 \geq \dots \geq \sigma_d \geq 0$ . Then

$$\|V^\top B_H\|_F^2 = \|V^\top B_{\text{res}}\|_F^2 = \sum_{i=1}^d \sigma_i^2 \|V^\top q_i\|_2^2.$$

Set  $a_i = \|V^\top q_i\|_2^2$ . Since  $V^\top V = I_k$  and  $Q$  is orthogonal, we have  $0 \leq a_i < 1$ ,  $\sum_{i=1}^d a_i = \text{tr}(V^\top Q Q V) = k$ . Because the singular values are ordered decreasingly, the weighted sum is maximized by assigning unit weight to its first  $k$  terms:

$$\sum_{i=1}^d \sigma_i^2 a_i \leq \sum_{i=1}^k \sigma_i^2.$$

Choosing  $V = U = [q_1, \dots, q_k]$  attains this bound. Moreover,  $U^\top B_H = U^\top B_{\text{res}}$ , and therefore

$$\|V^\top B_H\|_F^2 \leq \|U^\top B_H\|_F^2 = \sum_{i=1}^k \sigma_i^2 (B_{\text{res}}).$$

This completes the proof.  $\square$

### A.2 PROOF OF PROPOSITION 2

Let  $Q_H = (T_\Pi)_\# \tilde{P}_H$  and  $Q_F = (T_\Pi)_\# \tilde{P}_F$ , where  $T_\Pi(z) = \Pi z$ . Consider any coupling  $\gamma \in \Pi(Q_\Pi, \tilde{P}_F)$  and let  $(X, Y) \sim \gamma$ . The decomposition into the retained and removed subspaces gives

$$\begin{aligned} \|X - Y\|_2^2 &= \|\Pi(X - Y)\|_2^2 + \|(I_k - \Pi)(X - Y)\|_2^2 \\ &= \|\Pi(X - Y)\|_2^2 + \|(I_k - \Pi)Y\|_2^2. \end{aligned}$$

The joint distribution of  $(X, \Pi Y)$  is a coupling between  $Q_H$  and  $Q_F$ . Therefore,

$$\mathbb{E}_\gamma [\|X - \Pi Y\|_2^2] \geq W_2^2(Q_H, Q_F).$$

The second term depends only on the marginal distribution of  $Y$ , thus

$$\mathbb{E}_\gamma [\|(I_k - \Pi)Y\|_2^2] = \mathbb{E}_{Y \sim \tilde{P}_F} [\|(I_k - \Pi)Y\|_2^2].$$

It follows that

$$W_2^2(Q_H, \tilde{P}_F) \geq W_2^2(Q_H, Q_F) + \mathbb{E}_{Y \sim \tilde{P}_F} [\|(I_k - \Pi)Y\|_2^2].$$

To obtain the reverse inequality, let  $\eta$  be an optimal coupling between  $Q_H$  and  $Q_F$ . For each projected point  $u$ , sample  $Y$  according to the conditional distribution of  $\tilde{P}_F$  given  $\Pi Y = u$ . This lifts  $\eta$  to a coupling between  $Q_H$  and  $\tilde{P}_F$  for which  $(X, \Pi Y) \sim \eta$ . Under this coupling,

$$\begin{aligned}\mathbb{E}[\|X - Y\|_2^2] &= \mathbb{E}[\|X - \Pi Y\|_2^2] + \mathbb{E}[\|(I_k - \Pi)Y\|_2^2] \\ &= W_2^2(Q_H, Q_F) + \mathbb{E}_{Y \sim \tilde{P}_F}[\|(I_k - \Pi)Y\|_2^2].\end{aligned}$$

Combining the two inequalities gives

$$W_2^2(Q_H, \tilde{P}_F) = W_2^2(Q_H, Q_F) + \mathbb{E}_{Y \sim \tilde{P}_F}[\|(I_k - \Pi)Y\|_2^2].$$

Let  $\mu_F$  and  $\Sigma_F$  denote the mean and covariance of  $\tilde{P}_F$ . Then,

$$\begin{aligned}\mathbb{E}[\|(I_k - \Pi)Y\|_2^2] &= \|(I_k - \Pi)\mu_F\|_2^2 + \text{tr}((I_k - \Pi)\Sigma_F(I_k - \Pi)) \\ &= \|(I_k - \Pi)\mu_F\|_2^2 + \text{tr}((I_k - \Pi)\Sigma_F).\end{aligned}$$

Hence,

$$W_2^2((T_\Pi)_\# \tilde{P}_H, \tilde{P}_F) = W_2^2((T_\Pi)_\# \tilde{P}_H, (T_\Pi)_\# \tilde{P}_F) + \|(I_k - \Pi)\mu_F\|_2^2 + \text{tr}((I_k - \Pi)\Sigma_F).$$

Thus, if the removed subspace contains nonzero faithful energy, the projected distribution remains at a strictly positive Wasserstein distance from  $\tilde{P}_F$ . This completes the proof.  $\square$

### A.3 PROOF OF PROPOSITION 3

Let  $X = Z - \mu_H$  and  $\delta = \mu_F - \mu_H$ . Then  $\mathbb{E}[X] = 0$ ,  $\text{Cov}(X) = \Sigma_H$ , and

$$T_A(Z) - Z = \delta + (A - I_k)X.$$

Therefore,

$$\begin{aligned}\mathbb{E}[\|T^*(Z) - Z\|_2^2] &= \|\delta\|_2^2 + \mathbb{E}[\|(A - I_k)X\|_2^2] \\ &= \|\delta\|_2^2 + \text{tr}[(A - I_k)\Sigma_H(A - I_k)^\top] \\ &= \|\mu_F - \mu_H\|_2^2 + \text{tr}(\Sigma_H) + \text{tr}(A\Sigma_H A^\top) - 2\text{tr}(A\Sigma_H) \\ &= \|\mu_F - \mu_H\|_2^2 + \text{tr}(\Sigma_H) + \text{tr}(\Sigma_F) - 2\text{tr}(A\Sigma_H).\end{aligned}$$

The first three terms are independent of  $A$ . Thus, minimizing the expected distortion is equivalent to solving

$$\max_{A: A\Sigma_H A^\top = \Sigma_F} \text{tr}(A\Sigma_H).$$

For any feasible  $A$ , define  $Q = \Sigma_F^{-\frac{1}{2}} A \Sigma_H^{\frac{1}{2}}$ . The feasibility constraint implies

$$QQ^\top = \Sigma_F^{-\frac{1}{2}} A \Sigma_H A^\top \Sigma_F^{-\frac{1}{2}} = I_k.$$

So  $Q$  is orthogonal. Moreover,  $A = \Sigma_F^{\frac{1}{2}} Q \Sigma_H^{-\frac{1}{2}}$  and thus,

$$\text{tr}(A\Sigma_H) = \text{tr}(Q \Sigma_H^{\frac{1}{2}} \Sigma_F^{\frac{1}{2}}).$$

Let  $M = \Sigma_H^{\frac{1}{2}} \Sigma_F^{\frac{1}{2}}$ . By the von Neumann trace inequality, for any orthogonal  $Q$ ,

$$\text{tr}(QM) \leq \|M\|_* = \text{tr}((MM^\top)^{\frac{1}{2}}).$$

Since  $MM^\top = \Sigma_H^{\frac{1}{2}} \Sigma_F \Sigma_H^{\frac{1}{2}}$ , we have

$$\text{tr}(A\Sigma_H) \leq \text{tr}\left[(\Sigma_H^{\frac{1}{2}} \Sigma_F \Sigma_H^{\frac{1}{2}})^{\frac{1}{2}}\right].$$

We define  $S = (\Sigma_H^{\frac{1}{2}} \Sigma_F \Sigma_H^{\frac{1}{2}})^{\frac{1}{2}}$ . The proposed matrix  $A^* = \Sigma_H^{-\frac{1}{2}} S \Sigma_H^{-\frac{1}{2}}$  is feasible. Furthermore,

$$\text{tr}(A^* \Sigma_H) = \text{tr}(\Sigma_H^{-\frac{1}{2}} S \Sigma_H^{\frac{1}{2}}) = \text{tr}(S),$$

Therefore,  $A^*$  attains the upper bound and therefore minimizes the expected distortion. To establish uniqueness, let  $M = U\Lambda V^\top$  be the singular value decomposition of  $M$ . Because  $\Sigma_H$  and  $\Sigma_F$  are positive definite, all singular values in  $\Lambda$  are strictly positive. For any orthogonal  $Q$ ,

$$\text{tr}(QM) = \text{tr}(V^\top Q U \Lambda) \leq \text{tr}(\Lambda).$$

Equality holds only when  $V^\top Q U = I_k$ . Therefore, the corresponding feasible  $A$  is unique. Since  $A^*$  attains this maximum, it is the unique minimizer. Consequently,  $T^*$  achieves the target first and second moments without requiring  $Z$  to be Gaussian. This completes the proof.  $\square$

## B IMPLEMENTATION AND EVALUATION DETAILS

**Backbones and Paired Calibration Data.** We evaluate ResOT on LLaVA-1.5-7B, InstructBLIP-Vicuna-7B, and mPLUG-Owl2-LLaMA2-7B. All models are evaluated with greedy decoding to avoid randomness in generation. For each backbone, we use the same 5000 calibration pairs from the training data released with LURE (Zhou et al., 2024). Each pair associates an image from the COCO 2014 training set with a faithful caption from LLaVA-Instruct-150K and a corresponding hallucinated caption generated by GPT-3.5. We separately feed the faithful and hallucinated image-caption inputs through the frozen LVLM and extract their final-position hidden states at the selected decoder layer. During inference, ResOT repairs the hidden state used to predict the next token at each decoding step.

**Evaluation Protocols.** For POPE, we evaluate on the random, popular, and adversarial splits, each containing 3000 samples. A prediction is parsed by checking whether the first generated word is “yes” or “no”, otherwise it is counted as unknown. For CHAIR, we evaluate object hallucination on the COCO 2014 validation images (Lin et al., 2014). Following prior work (Huang et al., 2024; Yang et al., 2025a), we use the subset of 500 images selected by OPERA for efficient evaluation. The captioning prompt is fixed as: *Please describe this image in detail.* For MME, we use the official yes/no evaluation setting. Each image is associated with two questions, and the final score combines question-level accuracy and image-level pair accuracy. For GPT-assisted open-ended evaluation, we use GPT-4o as an external judge to assess image description quality. Given an image, the question, and the model responses, GPT is asked to score both responses from 1 to 10 in terms of accuracy and detailedness.

**Baseline Implementation.** We compare with decoding-based methods, including OPERA, VCD, HALC, and LVLMs-Saliency, and representation-level methods, including ICT, VTI, VISTA, Nullu, HulluEdit and REVIS. For each baseline, we use the official code and the hyperparameters recommended in the corresponding paper. All methods use greedy decoding except OPERA, for which we follow the official beam search setting with five beams. Within each benchmark, we keep the LVLM backbone, prompt, and maximum generation length the same across all methods.

**Hyperparameter Settings.** ResOT uses two rank parameters. The rank  $c$  determines the number of dominant directions used to estimate the faithful and hallucinated subspaces, while  $k$  determines the dimension of the residual repair subspace. We set  $c = k$  and select this shared rank from  $\{8, 16, 32, 64, 128\}$ . The repair strength  $\alpha$  is selected from  $\{0.1, 0.2, \dots, 1.0\}$ . At each decoding step, repair is applied only to the final-position hidden state in decoder layers 16-31. We set the covariance shrinkage coefficient to  $\rho = 0.05$  and the numerical stability term to  $\epsilon = 10^{-6}$ . For each layer, the correction threshold is set to the 95th percentile of OT correction norms computed from the hallucinated calibration representations. These settings are used across all models and benchmarks.

## C ADDITIONAL EXPERIMENTAL RESULTS

### C.1 RESULTS ON ADDITIONAL BENCHMARKS

We further evaluate ResOT on AMBER, MMHal-Bench, and MM-Vet (Wang et al., 2023; Sun et al., 2024; Yu et al., 2023). AMBER provides fine-grained analysis of object hallucination, MMHal-Bench evaluates hallucination in open-ended responses, and MM-Vet measures general multimodal capability across six skills. For AMBER, we use its generative split with the official prompt *Describe this image.* and report CHAIR, Cover, Hal, and Cog using the official LLM-free evaluator. For MMHal-Bench, we answer all 96 questions and use GPT-4o with the official evaluation prompt to report the average score and hallucination rate. For MM-Vet, we answer all 218 questions using their original prompts and follow the official GPT-assisted evaluation protocol.

**Results on AMBER.** Table 5 shows that ResOT consistently reduces hallucination on AMBER. It achieves the lowest CHAIR, Hal, and Cog values across all three backbones, demonstrating fewer hallucinated object mentions, fewer responses containing hallucinations, and less hallucination caused by contextual associations. These gains are achieved while preserving comparable object coverage, with Cover remaining within 1.9 points of the best result for each backbone.

Table 5: AMBER results across three LVLM backbones. Lower CHAIR, Hal, and Cog indicate less hallucination, while higher Cover indicates better object coverage.

| Backbone     | Method   | CHAIR ↓    | Cover ↑     | Hal ↓       | Cog ↓      |
|--------------|----------|------------|-------------|-------------|------------|
| LLaVA-1.5    | Baseline | 7.6        | 48.7        | 35.4        | 4.2        |
|              | Nullu    | 7.6        | <b>49.2</b> | 37.8        | 3.7        |
|              | ResOT    | <b>6.0</b> | 47.8        | <b>25.1</b> | <b>2.4</b> |
| mPLUG-Owl2   | Baseline | 9.3        | <b>50.3</b> | 40.8        | 5.3        |
|              | Nullu    | 6.6        | 48.3        | 27.7        | 3.0        |
|              | ResOT    | <b>4.6</b> | 48.4        | <b>18.7</b> | <b>1.6</b> |
| InstructBLIP | Baseline | 7.5        | <b>53.8</b> | 36.3        | 3.8        |
|              | Nullu    | 8.6        | 51.3        | 33.9        | 3.6        |
|              | ResOT    | <b>6.4</b> | 52.9        | <b>31.1</b> | <b>3.2</b> |

Table 6: MMHal-Bench results across three LVLM backbones. Avg. denotes average score, and Hall. denotes hallucination rate.

| Method   | LLaVA-1.5     |               | mPLUG-Owl2    |               | InstructBLIP  |               |
|----------|---------------|---------------|---------------|---------------|---------------|---------------|
|          | Avg. ↑        | Hall. ↓       | Avg. ↑        | Hall. ↓       | Avg. ↑        | Hall. ↓       |
| Baseline | 2.1250        | 0.6458        | 1.9479        | 0.6667        | 1.8229        | 0.6458        |
| Nullu    | 2.1146        | 0.6562        | 1.9583        | <b>0.6458</b> | 1.6250        | 0.6979        |
| ResOT    | <b>2.2708</b> | <b>0.5938</b> | <b>2.0104</b> | <b>0.6458</b> | <b>1.9583</b> | <b>0.6146</b> |

**Results on MMHal-Bench.** Table 6 shows that ResOT achieves the highest average score across all three backbones while reducing hallucination over the corresponding baselines. It obtains the lowest hallucination rate on LLaVA-1.5 and InstructBLIP and ties the best result on mPLUG-Owl2, demonstrating improved open-ended response quality and faithfulness.

**Results on MM-Vet.** Table 7 shows that ResOT improves the overall MM-Vet score across all three backbones, with gains spanning multiple multimodal skills. Together with the hallucination results, these findings demonstrate that ResOT improves visual faithfulness while maintaining or enhancing general multimodal capability.

## C.2 COMPLEXITY ANALYSIS

ResOT estimates the residual bases and Gaussian transport map offline, while online inference only applies the transport maps to the current token representations. Let  $L$  denote the number of repaired layers. At each layer, projecting the hidden state into the residual subspace and reconstructing it in the original hidden space require two matrix vector multiplications, each with a cost of  $O(dk)$ . Applying the affine transport map costs  $O(k^2)$ , while correction clipping costs  $O(k)$ . The likelihood gate is computed once for each generated token at the first layer and is shared by all repaired layers. Computing the gate costs  $O(k^2)$ . Therefore, the total additional complexity for each generated token is  $O(M(dk + k^2) + k^2)$ . Since  $k \ll d$ , the dominant term is  $O(Mdk)$ . The persistent repair state requires  $O(M(dk + k^2))$  memory and introduces no trainable parameters.

As shown in Table 8, ResOT retains 81.24% of the Vanilla throughput and increases only 8.147 MiB of peak GPU memory. It also keeps the LVLM backbone weights unchanged. ResOT therefore has low deployment overhead and can be enabled or disabled without changing the model checkpoint. Nullu is faster, but it directly modifies the backbone weights. Returning to the original model requires restoring these weights. Its suppression operation may also remove useful representations and affect general multimodal performance.



Table 7: MM-Vet results across three LVLm backbones. We report scores on recognition (Rec.), OCR, knowledge (Know.), generation (Gen.), spatial reasoning (Spat.), math, and the overall total score. Higher scores indicate better performance.

| Backbone     | Method  | Rec.        | OCR         | Know.       | Gen.        | Spat.       | Math        | Total       |
|--------------|---------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| LLaVA        | Vanilla | 34.0        | 22.6        | 16.1        | <b>20.1</b> | 24.4        | <b>11.5</b> | 29.5        |
|              | Ours    | <b>34.6</b> | <b>23.2</b> | <b>16.3</b> | 19.4        | <b>27.1</b> | <b>11.5</b> | <b>31.0</b> |
| mPLUG-Owl2   | Vanilla | <b>38.7</b> | 27.0        | <b>23.8</b> | <b>25.4</b> | 29.5        | <b>3.8</b>  | 34.3        |
|              | Ours    | 38.2        | <b>28.2</b> | 23.6        | 24.9        | <b>30.9</b> | <b>3.8</b>  | <b>34.5</b> |
| InstructBLIP | Vanilla | 30.9        | 15.9        | 15.4        | 16.0        | 17.3        | <b>8.1</b>  | 25.8        |
|              | Ours    | <b>33.5</b> | <b>17.1</b> | <b>19.6</b> | <b>18.0</b> | <b>18.1</b> | 7.3         | <b>27.7</b> |

Table 8: Inference efficiency on LLaVA-1.5 with 128 generated tokens. Changes are computed against each method’s paired Vanilla run.

| Method       | Tokens/s (vs. Vanilla) $\uparrow$ | Peak Memory Increase (MiB) $\downarrow$ | Preserves LVLm Weights |
|--------------|-----------------------------------|-----------------------------------------|------------------------|
| Vanilla      | 49.326 (0.00%)                    | 0.000                                   | ✓                      |
| OPERA        | 0.752 (−97.87%)                   | 25219.631                               | ✓                      |
| VCD          | 17.462 (−52.64%)                  | 724.160                                 | ✓                      |
| HALC         | 1.858 (−95.21%)                   | 4728.521                                | ✓                      |
| ICT          | 24.285 (−39.46%)                  | 0.250                                   | ✓                      |
| VTI          | 29.940 (−24.77%)                  | 56.504                                  | ✓                      |
| VISTA        | 22.200 (−51.43%)                  | 3761.718                                | ✓                      |
| Nullu        | 45.850 (−1.64%)                   | 0.000                                   | ✗                      |
| HulluEdit    | 17.843 (−56.16%)                  | 112.595                                 | ✓                      |
| <b>ResOT</b> | <b>40.071 (−18.76%)</b>           | <b>8.147</b>                            | ✓                      |

## D SENSITIVITY AND ABLATION ANALYSIS

### D.1 QUANTITATIVE ANALYSIS OF REPRESENTATION

For all analyzes in this subsection, we use the first 4000 faithful-hallucinated pairs for fitting and reserve the remaining 1000 pairs for evaluation. All visualizations and quantitative results are computed on the held-out split.

**Visualization across Layers.** Figure 6 extends the visualization in Figure 4 to layers 16 and 24. Across all three layers, the residual coordinates reveal a clearer mismatch between faithful and hallucinated representations than the space after Nullu suppression. This consistent pattern shows that filtering out the dominant faithful structure concentrates hallucination-related deviations in the localized repair space. Gaussian OT then aligns the hallucinated residual distribution with the faithful distribution at each layer, confirming that the localization and repair pattern observed in the main text also holds at the other selected layers.

**Subspace Separability.** To quantify the localization achieved by subspace construction, we compare the separability of faithful and hallucinated representations in the space after Nullu suppression and the ResOT residual space before repair. As shown in Table 9, the residual space consistently yields higher Mahalanobis distance, LDA accuracy, and LDA AUC across all evaluated layers, demonstrating that it better isolates hallucination-related deviations.

**Center Alignment by Gaussian OT.** Table 10 further quantifies the repair effect. Faithful and hallucinated centers are clearly separated in the residual subspace before repair. After applying Gaussian OT, the distance between the faithful and repaired centers converges to near zero across all evaluated layers. This demonstrates that Gaussian OT closes the localized center mismatch through transport rather than hard suppression.

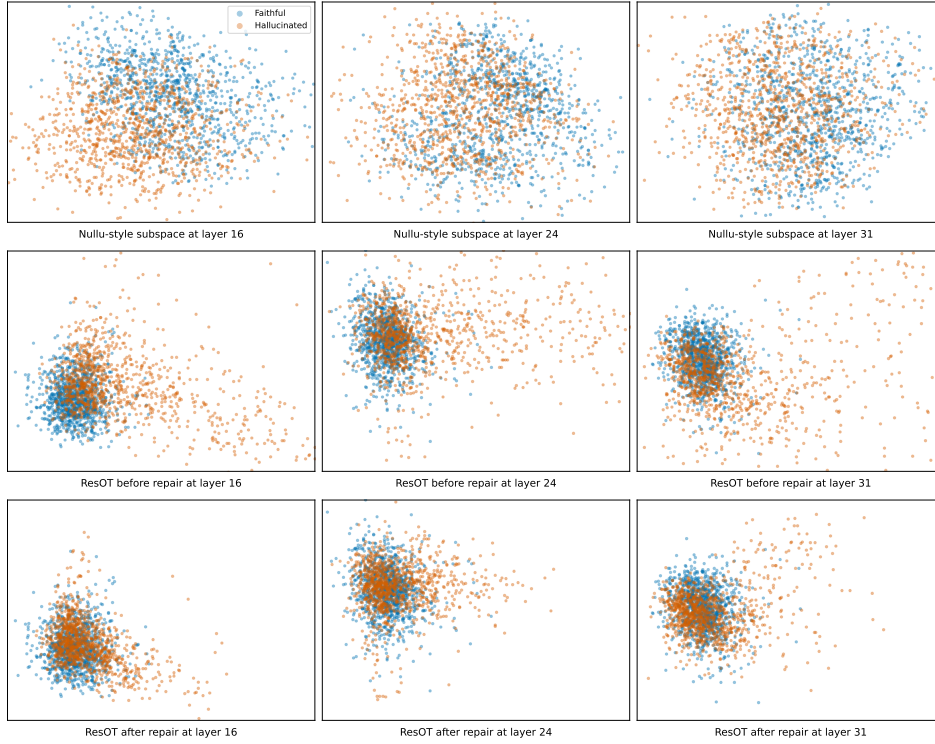


Figure 6: Multi-layer representation visualization at layers 16, 24, and 31. Rows correspond to the Nullu-style subspace, the ResOT residual subspace before repair and after repair.

Table 9: Separability between faithful and hallucinated LLaVA representations in the subspace on the evaluation split. Higher values indicate stronger separability.

| Layer | Setting             | Mahalanobis Dist. $\uparrow$ | LDA Acc. $\uparrow$ | LDA AUC $\uparrow$ |
|-------|---------------------|------------------------------|---------------------|--------------------|
| 16    | Nullu-style         | 1.5033                       | 0.7795              | 0.8674             |
|       | ResOT before repair | <b>2.4725</b>                | <b>0.8865</b>       | <b>0.9544</b>      |
| 24    | Nullu-style         | 1.3405                       | 0.7590              | 0.8366             |
|       | ResOT before repair | <b>2.1708</b>                | <b>0.8565</b>       | <b>0.9351</b>      |
| 31    | Nullu-style         | 1.2530                       | 0.7325              | 0.8161             |
|       | ResOT before repair | <b>1.9297</b>                | <b>0.8460</b>       | <b>0.9149</b>      |

**Selectivity of Representation Repair.** Table 11 shows that hallucinated representations consistently receive larger updates than faithful representations across layers. This pattern reflects the overall inference mechanism of ResOT, which combines OT correction within the residual subspace with adaptive control of repair strength. This demonstrates that ResOT concentrates changes on representations with stronger hallucination evidence rather than shifting all representations uniformly.

## D.2 HYPERPARAMETER SENSITIVITY

Figure 7 analyzes the sensitivity of ResOT to the repair strength  $\alpha$  and residual subspace rank on LLaVA-1.5. With the rank fixed at 32, increasing  $\alpha$  generally lowers  $C_s$  and improves BLEU, while MME peaks at  $\alpha = 0.8$  and drops beyond this point. Smaller ranks favor  $C_s$  and BLEU, whereas ranks above 32 weaken all three metrics. Overall, rank 32 and  $\alpha \in [0.7, 0.8]$  provide a favorable balance among hallucination reduction, captioning quality, and general multimodal capability.

Table 10: Center-distance analysis across layers.  $F$ ,  $H$ , and  $R$  denote faithful, hallucinated, and repaired representations, respectively.

| Layer | Nullu ( $F \rightarrow H$ ) | Before repair ( $F \rightarrow H$ ) | After repair ( $F \rightarrow R$ ) |
|-------|-----------------------------|-------------------------------------|------------------------------------|
| 16    | 1.9758                      | 1.2845                              | <b>0.0537</b>                      |
| 24    | 4.0293                      | 2.9075                              | <b>0.1265</b>                      |
| 31    | 4.4187                      | 2.5757                              | <b>0.0994</b>                      |

Table 11: Average relative displacement of faithful and hallucinated samples in the representation space. H/F denotes the ratio of hallucinated to faithful displacement. The 16–31 row reports the average across all selected layers.

| Layer | Faithful (%) | Hallucinated (%) | H/F   |
|-------|--------------|------------------|-------|
| 16    | 1.5903       | 7.1353           | 4.48× |
| 24    | 1.1953       | 5.9745           | 5.00× |
| 31    | 1.3840       | 4.2196           | 3.05× |
| 16–31 | 1.3425       | 6.0281           | 4.49× |

### D.3 GAUSSIAN DIAGNOSTICS IN THE HALLUCINATION RESIDUAL SUBSPACE

**Q1: Are the residual coordinates approximately Gaussian?** Figure 8 shows the diagnostics at layer 31. We first project faithful and hallucinated hidden states into the hallucination residual subspace. We then whiten each distribution using its estimated mean and covariance. The marginal histograms largely overlap with the standard normal reference. The marginal Q-Q plots also stay close to the diagonal in the central quantiles. Figure 9 shows a similar pattern across layers 16–31. The marginal KS distances are especially small for faithful representations. These results show that the Gaussian approximation describes the central region of individual coordinates reasonably well.

**Q2: Does marginal normality imply that the joint distribution is Gaussian?** No. The Mahalanobis Q-Q plots depart more clearly from the  $\chi_k^2$  reference. The corresponding KS distances are also larger. These differences are most visible in the tails. A coordinate can appear nearly Gaussian on its own even when the full distribution is not multivariate Gaussian. The normalized mean squared Mahalanobis distance,  $\mathbb{E}[d^2]/k$ , gives an additional check of the overall scale. However, a value near one does not prove joint Gaussianity. The Gaussian model is therefore a useful approximation of the first two moments, but it is not an exact model of the full distribution.

**Q3: Does the diagnostic examine the correction that is actually applied?** Yes. Rather than introducing an unrelated residual, we analyze the repair correction produced by the estimated OT map. For each hallucinated coordinate  $z_h$ , we compute

$$\delta_{\text{raw}} = T^*(z_h) - z_h, \quad \delta_{\text{clip}} = \min\left(1, \frac{\tau_{0.95}}{\|\delta_{\text{raw}}\|_2 + \epsilon}\right) \delta_{\text{raw}}, \quad (11)$$

where  $\tau_{0.95}$  is the 95th percentile of the correction norms from the same 5,000 calibration pairs used to fit the OT map. The final column of Figure 8 shows the whitened  $\delta_{\text{clip}}$ . At this stage, it has not been multiplied by the repair strength  $\alpha$  or the likelihood gate. Its central quantiles are approximately linear, while its tails depart from the Gaussian reference. The normalized mean squared Mahalanobis distance can fall below one. This occurs because norm clipping limits large correction norms and because the covariance used for whitening is regularized.

**Q4: Why is Gaussian OT still appropriate, and how are tail deviations handled?** Our method uses the Gaussian model to estimate transport from the mean and covariance. It does not assume that every projected activation is exactly Gaussian. The diagnostics show that this approximation captures the main structure in the central region. They also reveal deviations in the joint tails. We use covariance shrinkage to stabilize the estimated transport map. We then clip correction norms at the 95th percentile to bound unusually large interventions. These measures retain the simple Gaussian OT solution while limiting the influence of the observed tail deviations.

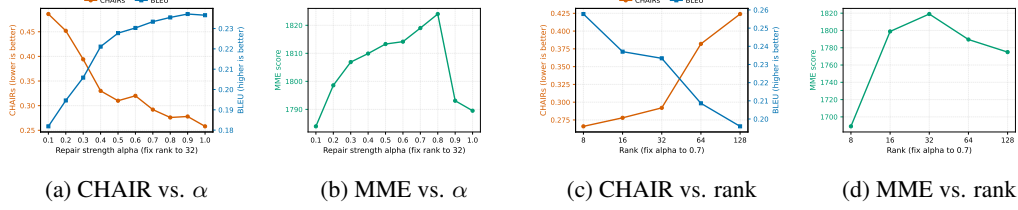


Figure 7: Sensitivity analysis of the repair strength  $\alpha$  and residual subspace rank on LLaVA-1.5. For  $\alpha$  sensitivity, we fix the rank to 32. For rank sensitivity, we fix  $\alpha$  to 0.7.

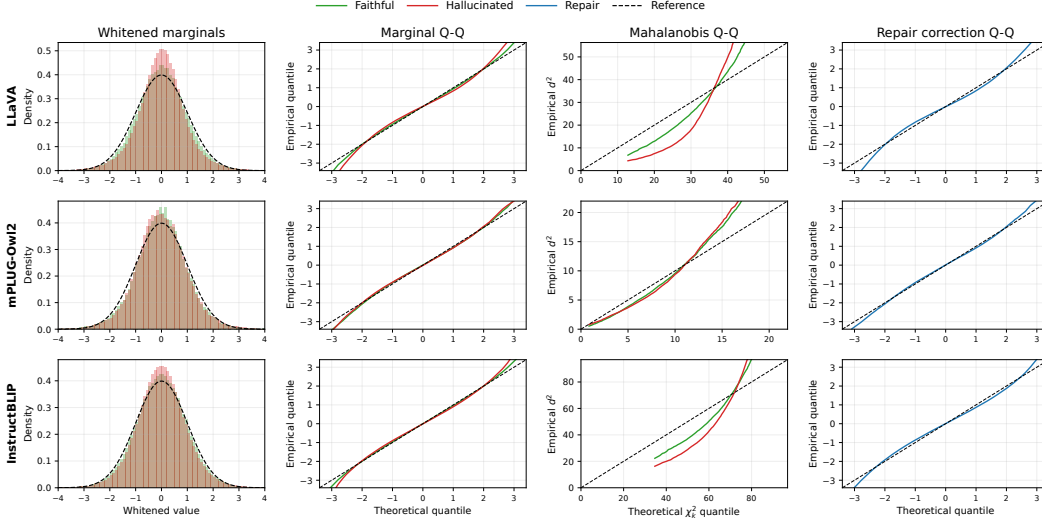


Figure 8: Gaussian diagnostics at layer 31 for LLaVA, mPLUG-Owl2 and InstructBLIP. The columns show whiten marginals, marginal Q-Q plots, Mahalanobis Q-Q plots and whiten repair corrections. The correction is  $T^*(z_h) - z_h$ . Its norm is clipped using the 95th percentile of the calibration correction norms. This is done before multiplication by the repair strength and the likelihood gate.

## E LIMITATION

ResOT intentionally uses a low dimensional Gaussian approximation to obtain a closed-form transport map. This design captures the main distribution shift through mean and covariance, rather than modeling every detail of the residual distribution. A useful direction is to explore richer transport families while retaining the current inference efficiency. In addition, our evaluation mainly focuses on object hallucination. Evaluating ResOT on other types of multimodal hallucination would provide a more complete understanding of its applicability.

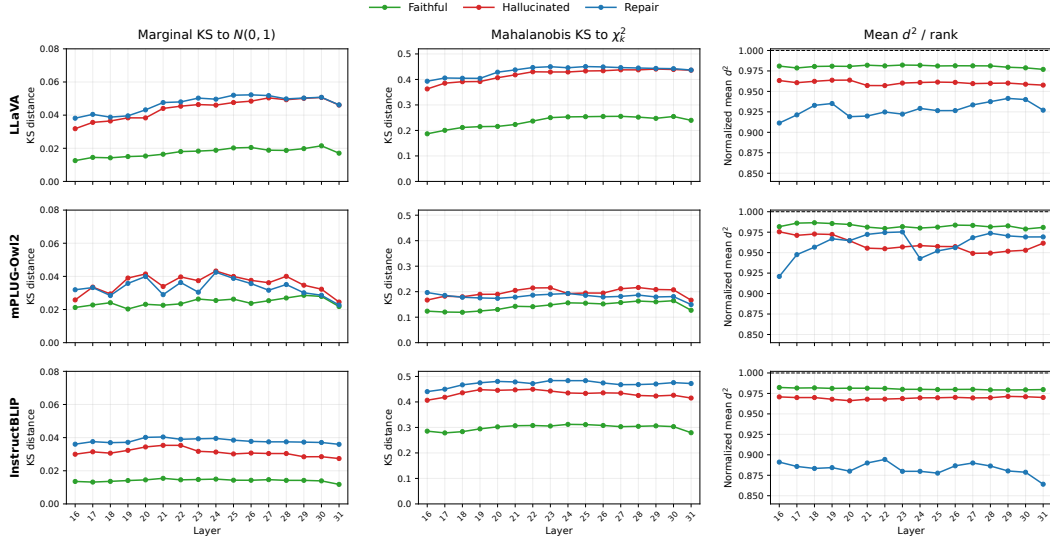


Figure 9: Gaussian diagnostics across layers 16–31 for LLaVA, mPLUG-Owl2 and InstructBLIP, using the same 5,000 calibration pairs as the OT maps. The columns report the marginal KS distance to  $N(0, 1)$ , the Mahalanobis KS distance to  $\chi_k^2$  and  $\mathbb{E}[d^2]/k$ . The curves compare faithful coordinates, hallucinated coordinates and repair corrections after norm clipping.